

Surface Integrals

$$z = g(x, y)$$

$$\iint_S \sqrt{1 + g_x^2 + g_y^2} dA$$

Goal:

To evaluate surface integrals and flux integrals

Definition: The Surface Integral of $f(x, y, z)$ over a surfaces, S

$$\iint_{S'} \underbrace{f(x, y, z)}_{g(x, y)} dS = \iint_R f(x, y, g(x, y)) \underbrace{\sqrt{1 + g_x^2 + g_y^2}}_{dS} dA$$

Note:

If $f(x, y, z) = \underline{1}$, then the integral gives the surface area of S on the domain.

Example:

Evaluate $\iint_S (x - 2y + \underbrace{z}_{g(x, y)}) dS$, where S is given by $z = 15 - 2x + 3y$ when $0 \leq x \leq 2$ and $0 \leq y \leq 4$.

$$\iint_{S'} f dS = \iint_R f \sqrt{1 + g_x^2 + g_y^2} dA$$

$$g_x = -2, g_y = 3$$

$$\int_0^2 \int_0^4 (x - 2y + \overbrace{(15 - 2x + 3y)}^z) \sqrt{1 + (-2)^2 + 3^2} dy dx$$

$$\int_0^2 \int_0^4 (-x + y + 15) \sqrt{14} dy dx$$

◦ exercise

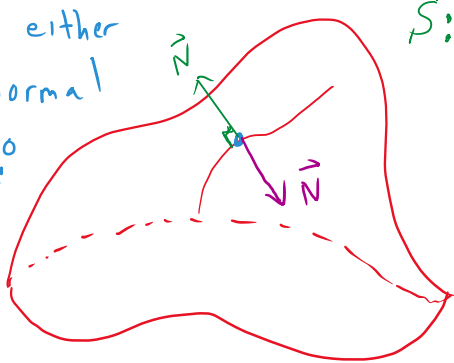
$$= 128\sqrt{14}$$

R : rectangle

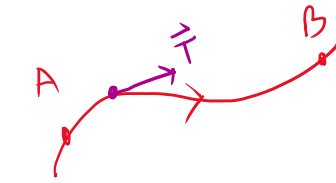
Surface Orientation

Two ways to orient a surface using a unit normal vector $\mathbf{N}$:

use either
unit normal
vector to
orient S



$$S: z = g(x, y)$$



For upward oriented $\vec{N}$, $\vec{k} \cdot \text{comp} > 0$
 " downward " " $\vec{k} \cdot \text{comp} < 0$

Notes:

1. Orient S by choosing an orientation for $\mathbf{N}$ (and stick with it).

2. Define $G(x, y, z) = z - g(x, y)$ and recall ∇G is

is normal to S (since S is a level surface of $G(x, y, z)$).

$$S: z = g(x, y)$$

$$z - g(x, y) = 0$$

$$G(x, y, z) = 0$$

3. The two choices for the unit normal vector are...

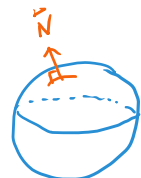
$$\vec{N} = \frac{\nabla G}{\|\nabla G\|} \quad \text{or} \quad \vec{N} = -\frac{\nabla G}{\|\nabla G\|}$$

4. For closed S , $\mathbf{N}$ always points ^{almost} outward

5. $\mathbf{N}$ must vary continuously for S to be orientable.

6. If S is defined parametrically by $\mathbf{r}(u, v)$, then

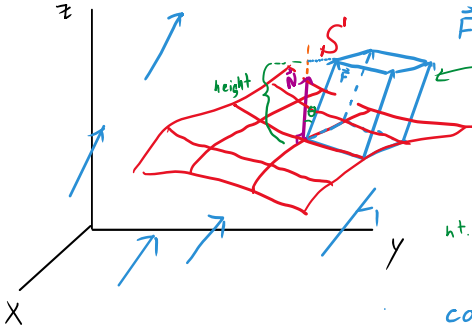
$$\vec{N} = \pm \frac{\vec{r}_u \times \vec{r}_v}{\|\vec{r}_u \times \vec{r}_v\|}$$



Flux Integrals

$$\vec{F} = \langle P, Q, R \rangle$$

Flux is the quantity of "fluid" passing through a surface per unit of time:



almost parallelepiped
so find volume of it.

$$(\text{Flux thru. surface patch}) \approx (\text{volume of parallelepiped})$$

$$= (\text{height}) (\text{Area of base surface patch})$$

$$\cos \theta = \frac{h}{\|\vec{F}\|}$$

$$= \|\vec{F}\| \|\vec{N}\| \cos \theta \, dS$$

$$= \underbrace{\vec{F} \cdot \vec{N}}_{\text{flux element}} dS \quad \text{Flux through a patch}$$

Definition: The Flux Integral of a Vector Field $\vec{F}$ Across a Surface S

$$\iint_{S'} \vec{F} \cdot \vec{N} \, dS$$

where $\vec{F} = \langle P, Q, R \rangle$ has continuous 1st partials

Notes

$$\textcircled{1} \quad \iint_S \vec{F} \cdot \vec{N} \, dS' = \iint_R \vec{F} \cdot (\pm \nabla G) \, dA \quad \leftarrow \text{calc. formula}$$

$\frac{\pm \nabla G}{\|\nabla G\|}$ $\sqrt{1 + G_x^2 + G_y^2} \, dA$

$\textcircled{2}$ If $\rho(x, y, z)$ is density of fluid at (x, y, z) (so $\vec{F}$ represents flow of fluid) then

$$\iint_S \rho(x, y, z) \underbrace{\vec{F} \cdot \vec{N} \, dS}_{\substack{\text{flux element} \\ = \text{volume}}} = \text{mass of fluid through } S, \text{ per unit of time}$$

mass element

(3) If S is given by $\vec{r}(u,v)$, then $\iint_S \vec{F} \cdot \vec{N} dS$
 $= \iint_D \vec{F} \cdot (\vec{r}_u \times \vec{r}_v) dA$

16.7
 (20) $\vec{F}(x,y,z) = \langle x, y, z \rangle$, S is the part of plane
 $S: x+y+z=2$ that lies in 1st octant; has upward
 $g(x,y) = 2-x-y$ orientation. Find flux through S .

$$\text{flux} = \iint_S \vec{F} \cdot \vec{N} dS = \iint_R \vec{F} \cdot \nabla G dA$$

$$\nabla G = 1\vec{i} + 1\vec{j} + 1\vec{k}$$

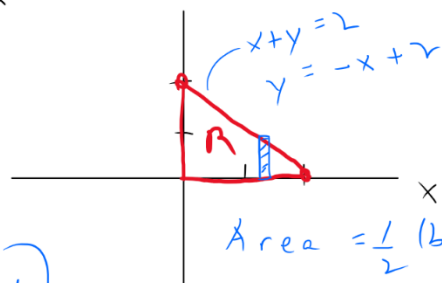
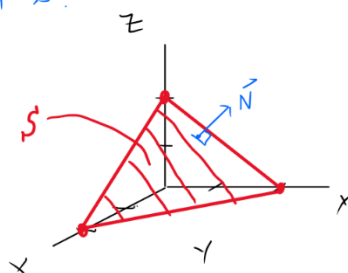
$$\vec{F} = \langle x, y, z \rangle$$

$$\nabla G = \langle 1, 1, 1 \rangle$$

$$\vec{F} \cdot \nabla G = x+y+z \quad \leftarrow 2-x-y$$

$$= 2$$

$$\text{flux} = \int_0^2 \int_0^{2-x} 2 dy dx = 2 \cdot 2 = (4)$$



$$\text{Area} = \frac{1}{2} (\text{base})(\text{height})$$

$$\frac{1}{2} \cdot 2 \cdot 2 = 2$$

Example:

Consider $\mathbf{F}(x, y, z) = (x + y)\mathbf{i} + y\mathbf{j} + z\mathbf{k}$, where S is the closed surface given by $z = 1 - x^2 - y^2$ and $z = 0$. Find $\iint_S \mathbf{F} \cdot \mathbf{N} \, dS$.